



BHOJ REDDY ENGINEERING COLLEGE FOR WOMEN
DEPARTMENT OF INFORMATION TECHNOLOGY
II B Tech. II Sem. A Section Project Abstract Seminar
Academic Year 2024-25 Date: - -2025

Student Council Election Hub

PRESENTED

BY

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INTRODUCTION

- In many educational institutions, students elections face challenges such as manual inefficiencies, lack of transparency, security concerns and low student engagement.
- Traditional voting methods, including paper ballots and in-person polling often lead to errors in vote counting, delay in results and announcements.
- The absence of a centralized, secure and transparent system results in distrust among students, discouraging participation in the democratic process.
- There is a need for digital, tamper proof election platform that ensures fairness, improves accessibility and provides real-time updates, making campus election more efficient and engaging.

EXISTING SYSTEM

- Existing systems like paper ballots, manual inefficiencies, security concerns and low student engagement.
- Votes are manually counted by election officials leads to human errors, voter fraud and delays in result announcements.
- In some institutions, it is difficult to engage with all the students, especially in large campuses.
- Lacks robust security measures (eg: multiple voting from the same user, unauthorized access).
- Voting locations are typically fixed, so only students who are on campus or near the polling stations can participate. This excludes students who may be off-campus are unable to physically get to a voting station.

DISADVANTAGES

- Lack of Transparency
- Security Risks
- Time – Consuming Process
- Low Student Engagement
- Campaigning pressure

PROPOSED SYSTEM

- The student council election hub is a secure, transparent and efficient digital voting platform designed to modernize student elections.
- This system eliminates the inefficiencies of traditional methods by integrating automated voting and strong security mechanisms.
- Students can cast votes online using a web or mobile platform.
- Multi-factor authentication(student ID) ensures only authorized students vote.
- Each candidate can create a profile with the information about themselves, their qualifications, goals and campaign promises.

ADVANTAGES

- Efficiency and Time-Saving
- Increased Student Participation
- Cost-Effective
- Sustainability
- Security & Data-Integrity

FUNCTIONAL REQUIREMENTS

These are the requirements that the end user specifically demands as basic facilities in the form of input to be given to the system, the operation performed and the output expected. They are basically the requirements stated by the user which one can see directly in the final product.

- User Management
- Vote Updates
- View Candidates rules, positions
- Cast your Vote
- Cloud Resource Management

NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the qualities or attributes that the system must have, such as performance, security, usability and scalability. It should deliver fast, seamless performance across devices. Compliance with security standards ensures data integrity, user privacy, and fraud prevention.

- Performance
- Security
- Scalability
- Usability
- Maintainability

MODULES

1. Student Management Module

- Student registration and login.
- Profile management.

2. Election Management Module

- Defines election rules, dates and positions.
- Review and approve candidate applications.

3. Candidate Promotion Module

- Candidates display their qualifications, goals and campaign promises.
- Allow students to engage with candidates.
- Students can submit questions to be answered by the candidates.

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4. Voting & Authentication Module

- Multi-factor authentication (student ID verification, login) is provided.
- Ensures privacy and prevents duplicate voting.

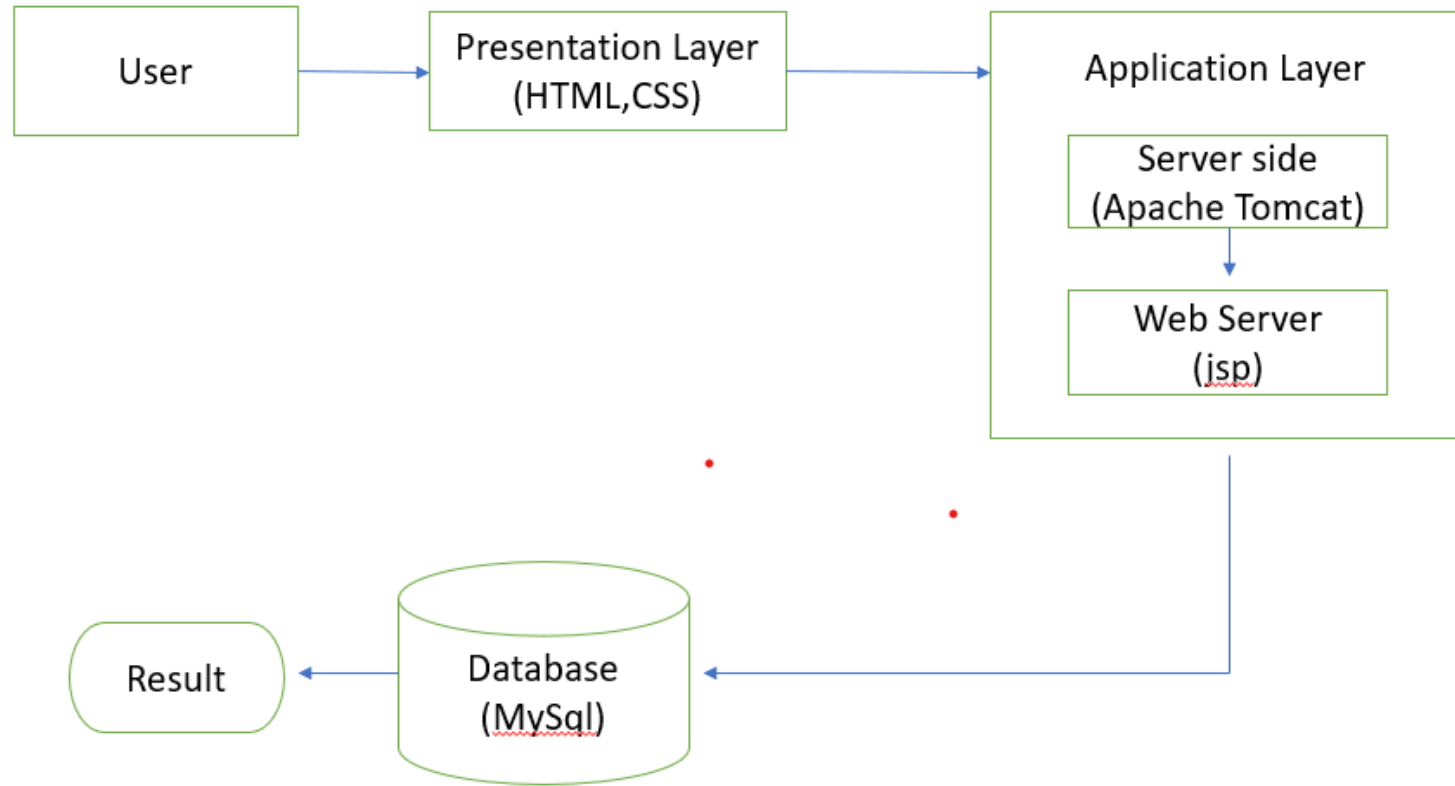
5. Cloud Resource Management Module

- Ensures scalability and optimal resource utilization.
- Load balancing to handle large user volumes.

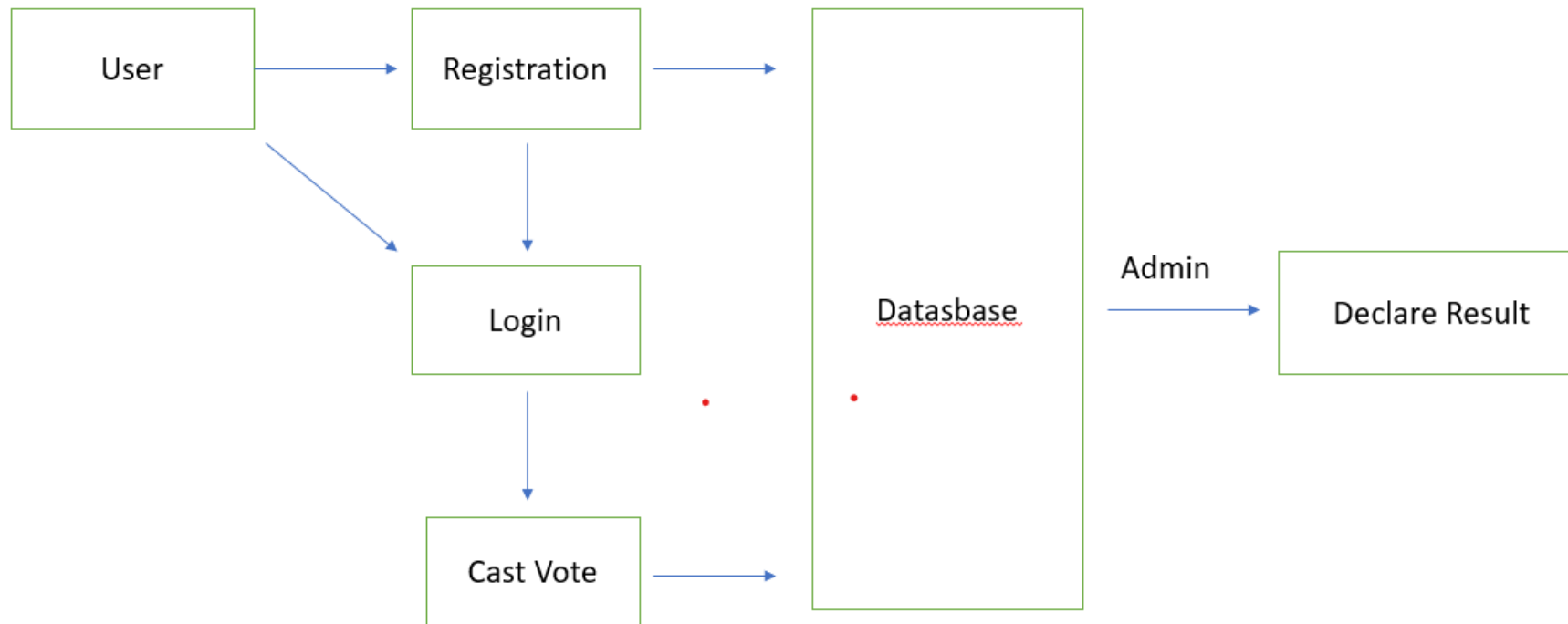
6. Security and Fraud Prevention Module

- Safeguards the system from unauthorized access and fraud.
- User authentication.

TECHNICAL ARCHITECTURE



SYSTEM ARCHITECTURE



COMPUTATIONAL RESOURCES

Software Requirements:

- Front-end : HTML, CSS, JavaScript.
- Back-end : Node JS, Python.
- Database Management : MySQL.
- Development Tools : Visual Studio Code.

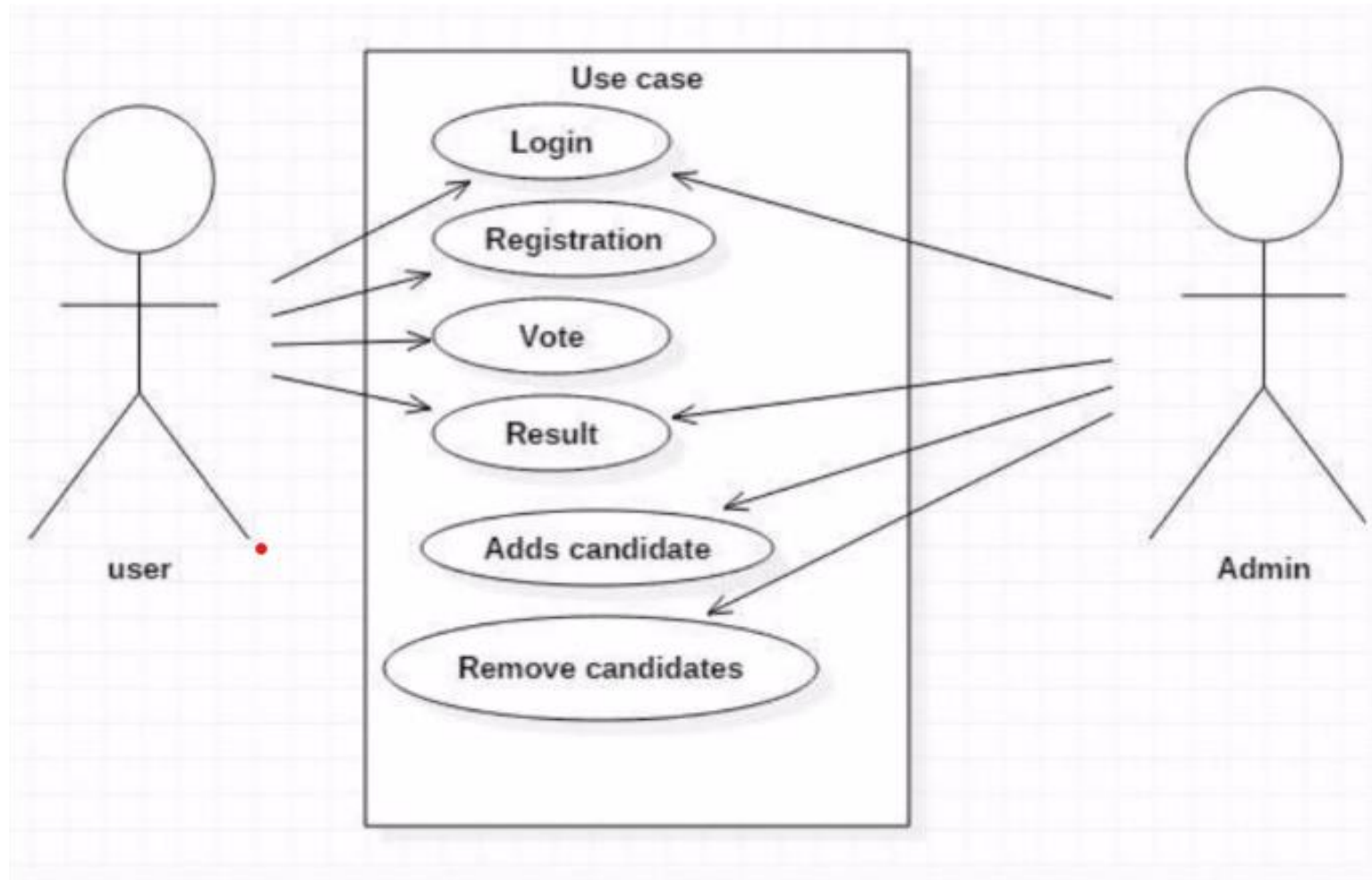
COMPUTATIONAL RESOURCES

Hardware Requirements:

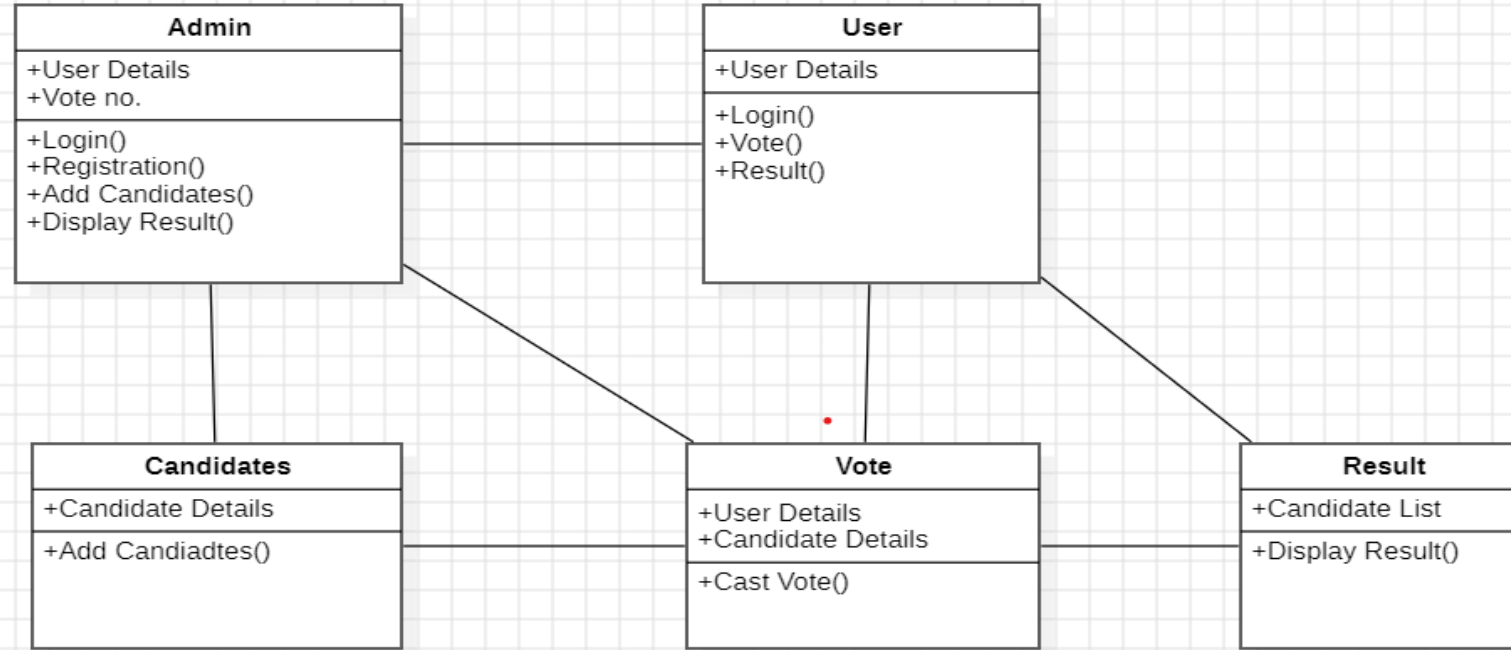
- Processor : Multi-core CPU (e.g., Intel Xenon).
- RAM : 8GB or above.
- Hard Disk : 100 GB or more.
- Operating System : Windows 10, or Linux

UML DIAGRAMS

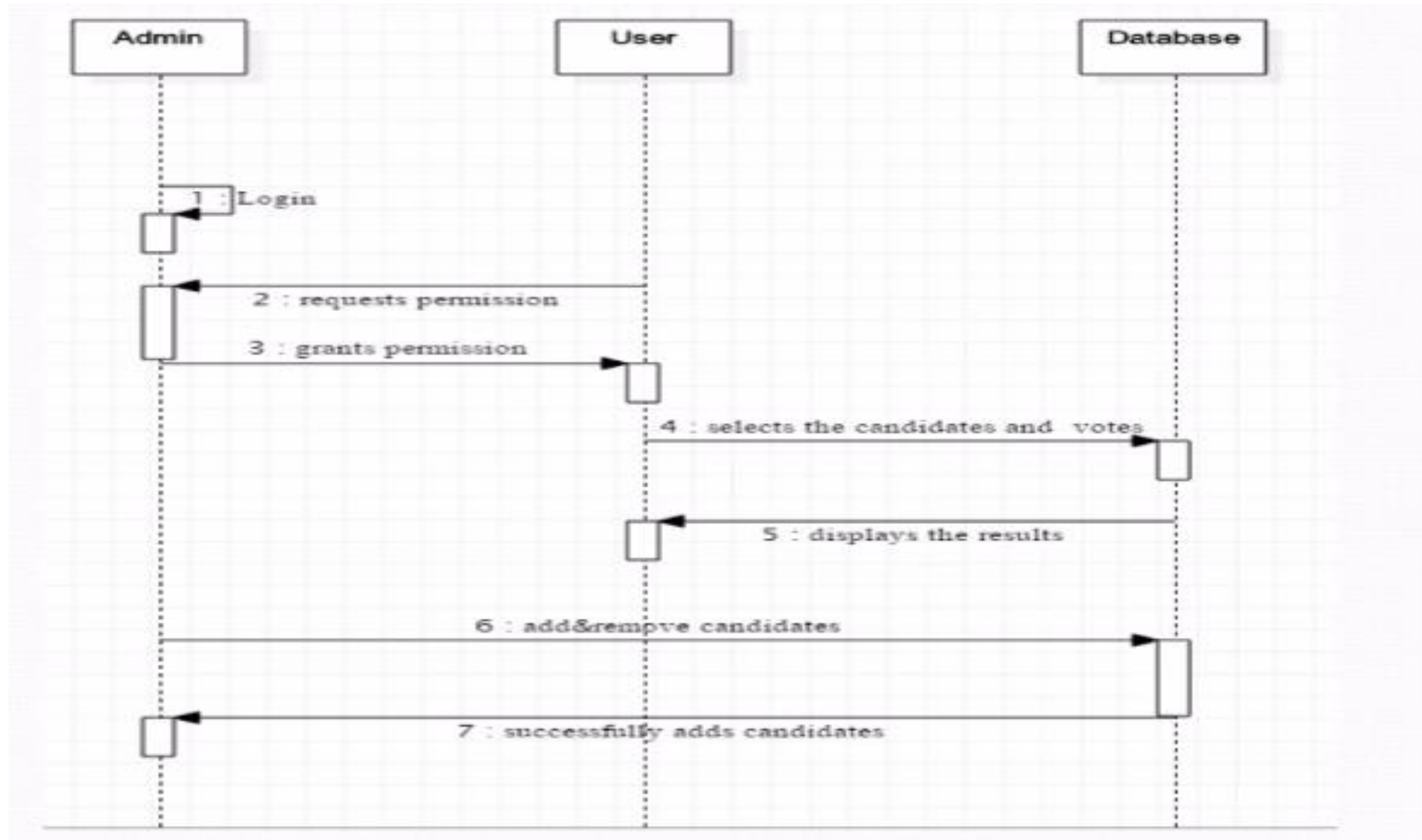
- Use case diagram:



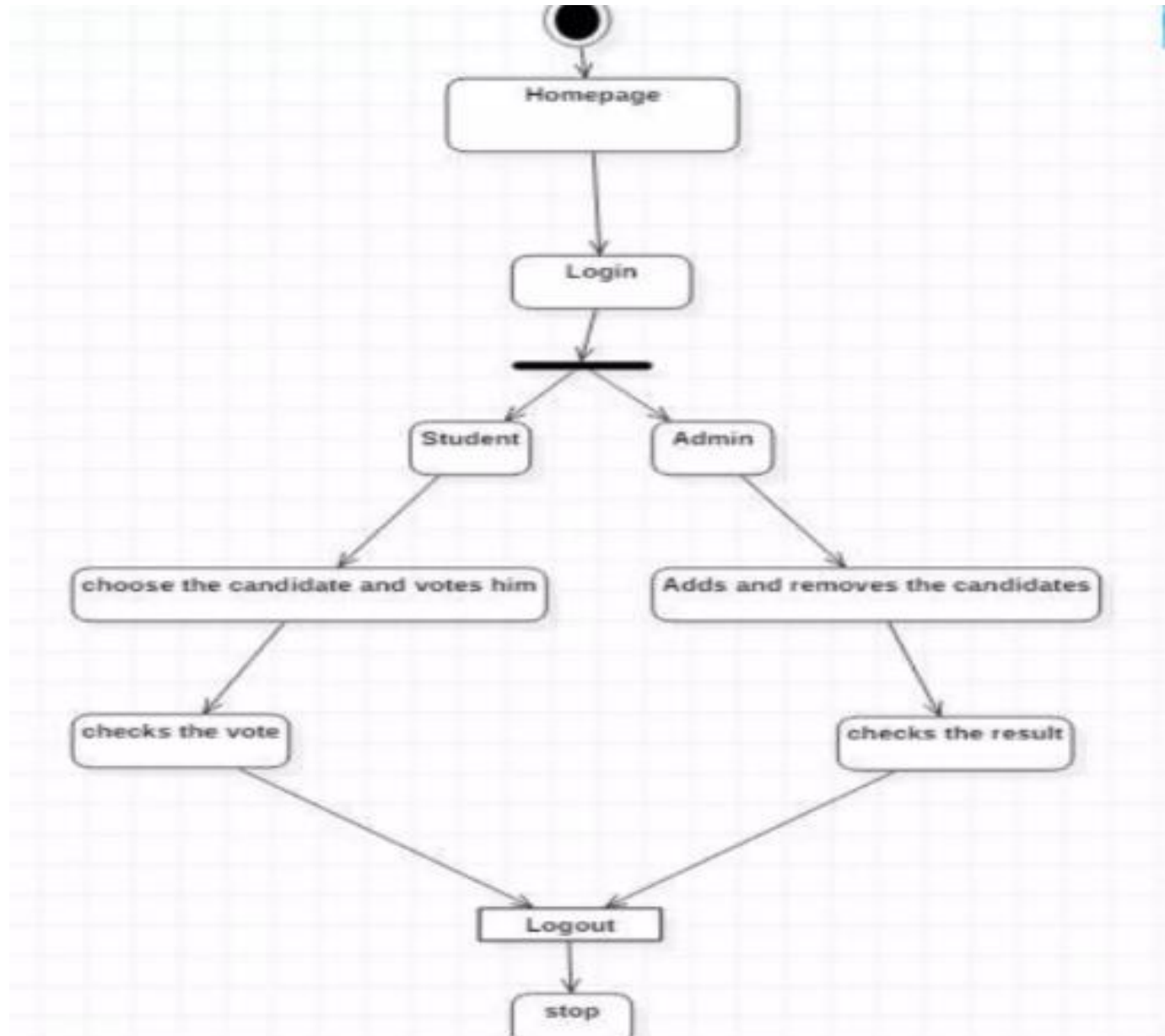
Class diagram:



Sequence diagram:



Activity diagram:



Implementation:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>student election</title>
<link rel="icon" type="image" href="logo.1.jpg">
<link rel="stylesheet" href="style1.css">
</head>
<body>
<section class="header">
<nav>
<header class="election-header">
<h1>Student Council Election Hub</h1>
<p class="tagline">Your Vote. Your Voice. Your Future.</p>
</header>
<div class="nav-links">
<ul>
<li><a href="main.html">Home</a></li>
```

```
<li><a href="About.html">About</a></li>
<li><a href="register.html">Register</a></li>
<li><a href="login.html">Login</a></li>
<li><a href="add_candidate.jsp">candidate</a></li>
<li><a href="vote.jsp">vote</a></li>
</ul>
</div>
</nav>
</section>
<div class="f1">
<h1 style="text-align:center">ONLINE VOTING SYSTEM</h1><br>
<br>
<p align="center" >An online voting system is a software platform that allows groups to securely conduct votes and elections. High-
quality online voting systems balance ballot security, accessibility, and the overall requirements of an organization's voting event.
This system typically aims to modernize the way student government elections, club votes, or any other campus-related decisions are
made</p>
</div>
<section class="Key Features">
<h1 align="center">Key Features of a Student Online Voting System:</h1>
<div class="row">
<div class="user">
```

User Authentication

Students can log in to the system using their university credentials (like student IDs and passwords). Authentication ensures that only eligible students can vote.

Election Creation

Administrators (like the student council or election organizers) can create and manage elections, including defining the candidates, voting options, and election period.

Voting Process

Once the election starts, students can view the list of candidates and their respective platforms, or vote on specific issues (e.g., for student body president, class representatives, etc.).

Voting can be done through a simple interface with radio buttons, checkboxes, or dropdown menus.

Security and Integrity

The system must ensure the integrity of the voting process by preventing duplicate voting (each student can vote only once). Secure encryption methods should be used to protect voting data and student privacy.

```
</div>
```

```
<div class="user1">
```

```
<h2>Mobile and accessibility</h2>
```

```
<p>Students can vote from their computers, tablets, or mobile phones, making the process more  
convenient and accessible.</p>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<br>
```

```
</body>
```

```
</html>
```

```
</p>
```

```
*{
```

```
margin: 0;
```

```
padding: 0;
```

```
}
```

```
.header{
```

```
height:950px;
```

```
width:1500px;
```

```
background-image:url("image3.1.jpg");
```

```
background-position: center;
```

```
background-size:cover;
```

```
background-repeat:no-repeat;
```

```
}
```

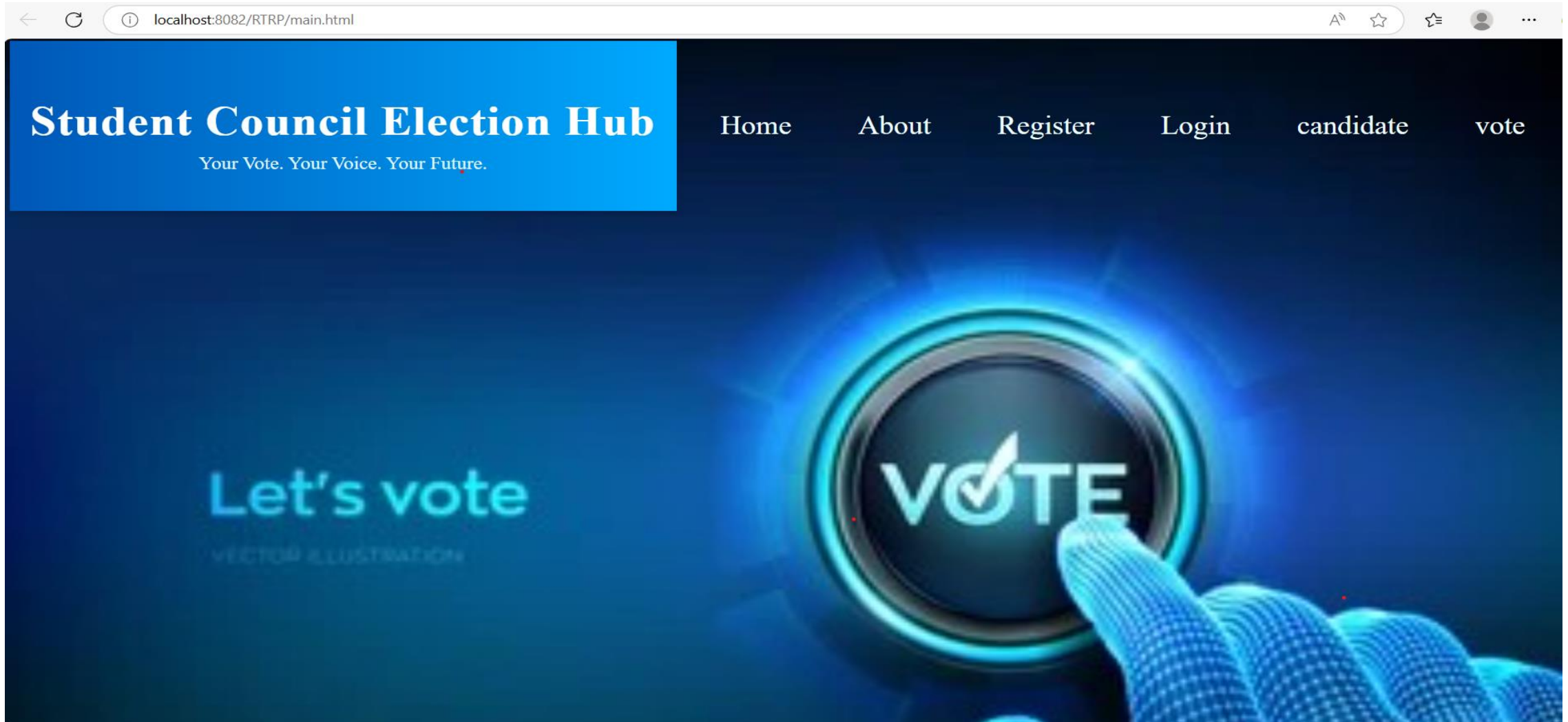
```
nav {  
  display: flex;  
  padding: 2px 6px;  
  justify-content: space-between;  
  align-items: center;  
}  
nav img{  
  width:150px;  
}  
.nav-links{  
  flex: auto;  
  text-align: right;  
}  
.nav-links ul li{  
  list-style: none;  
  display:inline-block;  
  padding:30px;  
}  
.nav-links ul li a{  
  font-size: 28px;  
  color: azure;  
  text-decoration: none;  
}
```

```
.fl {  
font-family: Arial, sans-serif ;  
}  
* {  
margin: 0;  
padding: 0;  
box-sizing: border-box;  
}  
.fl {  
padding:50px;  
}  
.fl p{  
font-family:'Times New Roman', Times, serif;  
}  
.row{  
margin-top:5%;  
display:flex;  
justify-content: space-between;  
text-align: center;  
gap: 50px;  
.user{  
flex: basis 31;  
background-color: rgb(31, 195, 228);  
border-radius: 10px;
```

```
margin-bottom: 5%;
padding: 20px 12px;
box-sizing: border-box;
text-align: center;
}
.col{
margin-top: 5%;
display: flex;
justify-content: space-between;
text-align: center;
gap: 5%;
}
.user1 {
flex: basis 31;
background-color: rgb(31, 195, 228);
border-radius: 10px;
margin-bottom: 5%;
padding: 20px 12px;
box-sizing: border-box;
text-align: center;
}
.user1 p{
font-family: 'Times New Roman', Times, serif;
}
```

```
nav h1 {  
  font-size: 80px;  
}  
.election-header {  
  background: linear-gradient(to right, #0057b7, #00aaff);  
  color: white;  
  text-align: center;  
  padding: 60px 20px 40px;  
  box-shadow: 0 4px 8px rgba(0,0,0,0.1);  
}  
.election-header h1 {  
  margin: 0;  
  font-size: 2.8rem;  
  letter-spacing: 1px;  
}  
.election-header .tagline {  
  font-size: 1.2rem;  
  margin-top: 10px;  
  color: #e0f7ff;  
}
```

Screenshots:



Student Council Election Hub

Login to vote and track results!

firstname

laxmi

Password

••••••••

[Forgot password?](#)

Login

Don't have an account? [Register here](#)



localhost:8082/RTRP/register.html



Student Council Election Hub - Register

Already have an account? [Login here](#)

Add New Candidate

20

Rajamouli

Add Candidate

Vote for Your Candidate

Select	Candidate	Votes
☐	Dommeti LaxmiPrasanna	0
☐	Poreddy LaxmiSree	1
☐	Mokurala Ganesh Goud	2
☐	Mokurala Lavanya	0
☐	Dommeti Sowmya	3
☐	vaddepally Lahari	0

Vote Now

View Results

Voting Results

Candidate	Votes
Dommeti Sowmya	3
Mokurala Ganesh Goud	2
Poreddy LaxmiSree	1
Dommeti LaxmiPrasanna	0
Mokurala Lavanya	0
vaddepally Lahari	0



Empowering Students, Strengthening Democracy

The *Student Council Election Hub* is an online platform designed to streamline student elections, ensuring fairness, security, and accessibility. Our goal is to provide a *transparent, efficient, and user-friendly* voting system for students.

Features of Student Council Election Hub:

- Secure online voting with encrypted authentication
- Live election results & real-time updates
- User-friendly dashboard for voters and administrators
- Multiple election categories (President, Secretary, etc.)

```
mysql> select*from candidates;
```

id	name	votes
1	Dommeti LaxmiPrasanna	0
2	Poreddy LaxmiSree	1
3	Mokurala Ganesh Goud	2
4	Mokurala Lavanya	0
5	Dommeti Sowmya	3
8	vaddepally Lahari	0

```
6 rows in set (0.00 sec)
```

```
mysql> select*from users;
```

id	firstname	lastname	department	password
1	mahi	kadari	it	123456789
2	laxmi	prasanna	cse	12345678
4	madhu	kadari	cse	1234567890

```
3 rows in set (0.00 sec)
```

CONCLUSION:

- The Student Council Election Hub serves as a vital platform that promotes democratic participation, transparency, and student engagement within the college community.
- By providing easy access to candidate information, voting procedures, and election updates, the hub empowers students to make informed decisions and encourages active involvement in college governance.
- Ultimately, it fosters leadership skills and a sense of responsibility among students, contributing to a more vibrant and attractive environment.

FUTURE SCOPE

- Integrate real-time voting analytics
- Add multimedia candidate profiles (videos, speeches, manifestos)
- Enhanced Security
- Implement mobile app compatibility for easier access and voting
- Also tracking results successfully

References:

- Open Source Voting System (GitHub)
- Online Voting System (using MySQL)
- Edutopia – Student Council Election Tips

Thank
you

